

Dong Payton Pei

Ottawa, ON | paytonpei01@gmail.com | github.com/mammut001 | www.linkedin.com/in/pd110

SUMMARY

Software developer and Master of Engineering student in System Science and Engineering at the University of Ottawa. Former Nokia and Ford co-op developer/tester with experience building full-stack, desktop, mobile, and AI products using TypeScript, React, Rust/Tauri, Swift, Kotlin, and cloud APIs.

EDUCATION

Carleton University <i>Bachelor of Computer Science</i>	Ottawa, ON 2017 — 2023
University of Ottawa <i>Master of Engineering in System Science and Engineering</i>	Ottawa, ON 2024

WORK EXPERIENCE

Software Developer Co-op Ford Motor Canada	Apr 2020 — Aug 2020 Ottawa, ON
- Developed C++11 unit-test scripts with GoogleTest for automotive software components. - Integrated test execution with Jenkins CI/CD in an Agile development workflow.	
Software Tester Co-op Nokia Canada	Aug 2020 — Aug 2021 Ottawa, ON
- Improved JavaScript/XML test code to strengthen coverage, maintainability, and regression confidence. - Documented system architecture and data flows in Confluence to support cross-team development.	

PROJECTS

pipi-shrimp-agent	github.com/mammut001/pipi-shrimp-agent
Built a local-first AI desktop agent with Rust, Tauri 2, React, and TypeScript, supporting multi-provider streaming, tool calling, workflow orchestration, and project-scoped memory.	
Resume Generator	resume-tailor.paytonpei.top
Built and deployed a full-stack resume editor with React, TypeScript, Vite, and Node.	
Focus Mint	apps.apple.com/us/app/focus-mint-focus-timer-study/id6759029810
Shipped a SwiftUI productivity app for iOS, watchOS, and macOS with Pomodoro sessions, widgets, multilingual localization, in-app purchases, and cross-device synchronization.	
Meet See U	chatpart-website.vercel.app
Built a native Android AI companion app with Kotlin and Jetpack Compose, featuring Firebase authentication, customizable character profiles, messaging, push-to-talk/VAD voice chat, semantic memory, and Stripe-paid character slots.	
Conveyor	github.com/mammut001/Conveyor
Built a Python automation platform for Telegram and Feishu that routes commands to agent-assisted development tasks, personal tools, and scheduled operations.	

RESEARCH

Low-Light Image Enhancement with DDBM	2024 — Present
Conducting research on low-light image enhancement with DDBM, building on the Learning to See in the Dark research direction.	

SKILLS

- Languages:** TypeScript, JavaScript, Python, Java, C++, Swift, Kotlin, Rust, HTML, CSS, Bash
- Frameworks & Data:** React, React Native, Next.js, SwiftUI, Jetpack Compose, Node.js, Tailwind CSS, Flask, Tauri 2, SQLite, MongoDB, Firebase, WebRTC, PyTorch, Typst, Mermaid
- Testing & Delivery:** GoogleTest, PyTest, SeleniumGrid, Jenkins, Docker